

Hao Ke

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EDUCATION

Beijing Jiaotong University,

M.S. in Information and Communication Engineering

• GPA 3.4/4.0 Academic Ranking: 5/31

• Advisors: Prof. Wei and Prof. Chang

Beijing, China

2024 – Present

China Three Gorges University

B.S. in Communication Engineering

• GPA 3.0/4.0

• CET4 500/710 CET6 451/710 IELTS 6.5/9.0

Yichang, Hubei, China

2019 – 2023

EXPERIENCE

M.S., Computer Science and Technology, Beijing Jiaotong University

2024 – Present

- Research on 3D reconstruction and **neural rendering** for **transparent and reflective scenes**.
- Research on consistent **3D Gaussian** scene **update** and 3D scene **editing**.
- Research on geometrically consistent **feed-forward** 3D reconstruction with 3D Gaussian Splatting.
- Project: Cross-modal generation technology for 3D server room spaces.
- Project: Based on 3D large-model digital space synthesis technology
- Master's Thesis Topic: High-quality 3D reconstruction method for complex media and occluded scenes.

Algorithm Engineer(Intern), Automotive and Robotics, Xiaomi Corporation

2026.01 – 2026.10

- Research on Action-Conditional **world models** for future frame prediction and dynamic scene modeling.
- Research on training **vision-language models** for monocular depth estimation and 3D scene understanding.
- Research on **3D occupancy prediction** and occupancy-grounded VLMs for 3D scene understanding.
- Project: Leveraged V-JEPA 2.1 for spatiotemporal representation learning from multi-view driving scenes.

PUBLICATIONS

Glass-Casing-Free 3D Reconstruction, *IEEE Transactions on Multimedia*

2026

- For scenarios involving complex media such as glass that are reflective or semi-transparent.
- A 3D reconstruction method based on implicit representation.

Consistent 3D Gaussian Scene Update via Voxel-Based Localization, (*Under Review*)

2026

- Proposed voxel-based ray sampling for 3D change localization with reduced sensitivity to primitive distribution.
- A voxel-guided 3D Gaussian Splatting (3DGS) scene update method.

COMPETITIONS AND PATENTS

- Patent: A 3D reconstruction method for transparent casing removal. 202610480833.5 2026
- World Models Track of the GigaBrain Challenge 2026 CVPR Workshop. (Achieved 3rd place) 2026
- The 5th National Undergraduate Embedded Chip and System Design Competition - FPGA Innovation Design Competition. (First Prize) 2021

AWARDS

- Graduate Academic Scholarship, Beijing Jiaotong University
- Outstanding Graduate, China Three Gorges University